

Vector Electronics

QuBi

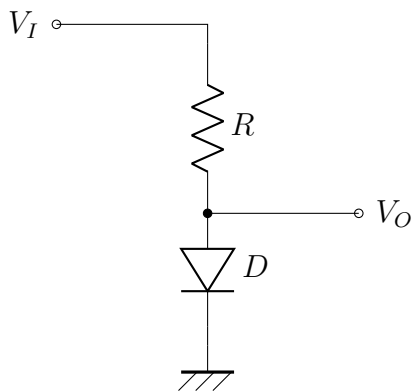
October 27, 2025

The purpose of this article is to describe a method for solving electronic circuits involving non-linear devices like diodes, bipolar transistors, MOSFETs, etc.

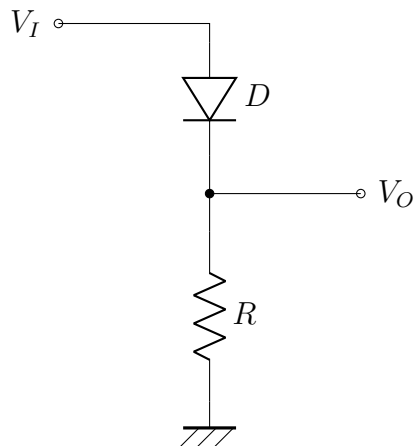
An extension of the Small Signal Analysis

In the following, we will consider that the non-linear device has a characteristic curve that can be modelled as a *piecewise linear function*. From that point on, we will investigate where this leads us.

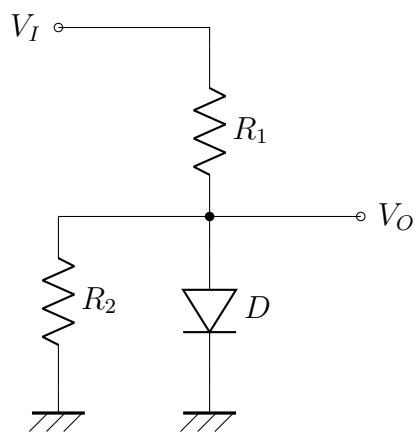
Example 1: diode and resistor I



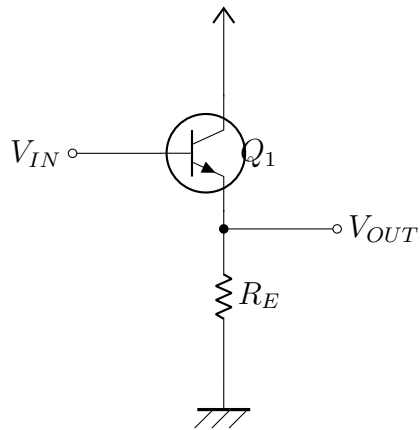
Example 2: diode and resistor II



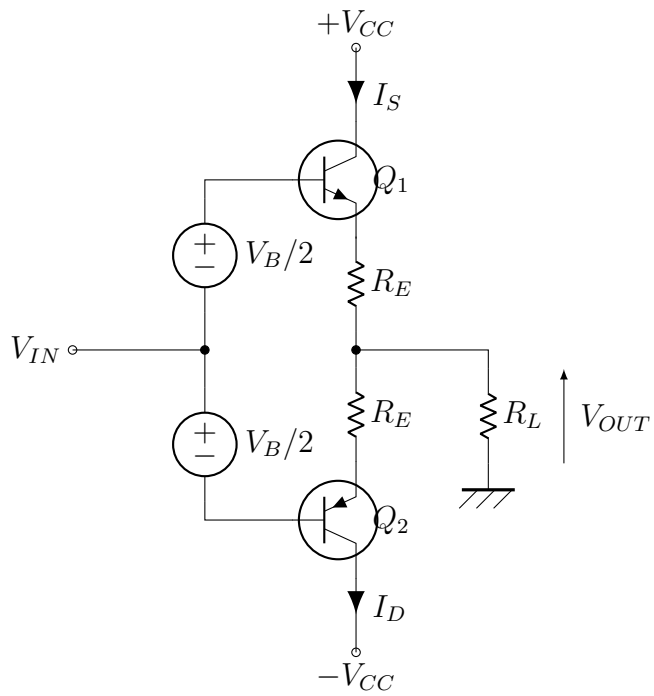
Example 3: 1 diode and 2 resistors



Example 4: common emitter



Example 5: class AB amplifier



Derivation Assuming that $I_C = f_S(V_{BE})$ for the NPN and $I_C = f_D(V_{EB})$ for the PNP we get the following:

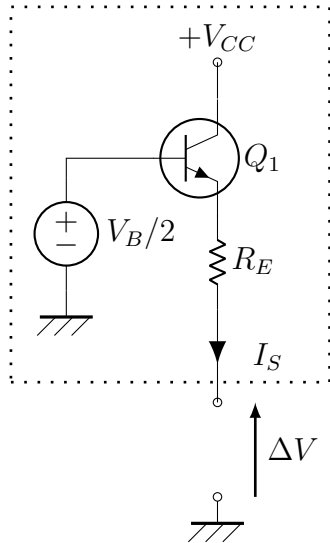
$$\begin{cases} I_S = f_S \left(v_{IN} - v_{OUT} + \frac{v_B}{2} - R_E I_S \right) \\ I_D = f_D \left(-(v_{OUT} - v_{IN}) + \frac{v_B}{2} - R_E I_D \right) \\ V_{OUT} = R_L(I_S - I_D) \end{cases}$$

With a linear assumption, it becomes:

$$\begin{cases} I_S = \frac{a_S}{1 + R_E a_S} (V_{IN} - V_{OUT} + \frac{V_B}{2}) + \frac{b_S}{1 + R_E a_S} \\ I_D = \frac{a_D}{1 + R_E a_D} (-(V_{IN} - V_{OUT}) + \frac{V_B}{2}) + \frac{b_D}{1 + R_E a_D} \\ V_{OUT} = R_L(I_S - I_D) \end{cases}$$

This circuit is a bit trickier. Let's break it down as 2 separate components.

Class AB subsystem: the current source



Voltage and current law give the following:

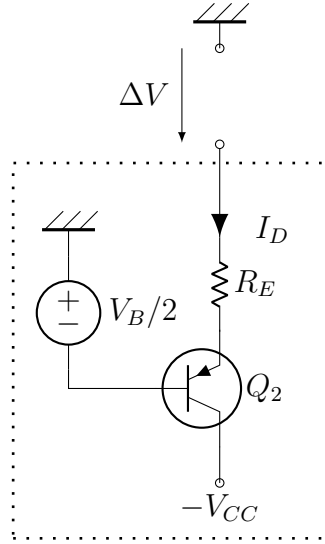
$$I_S = f_S \left(-\Delta V + \frac{v_B}{2} - R_E I_S \right)$$

Under a linear model assumption $f_S(V) = aV + b$, we can write that:

$$I_S = a \left(-\Delta V + \frac{v_B}{2} - R_E I_S \right) + b$$

Finally, after a bit of algebra:
$$I_S = -\frac{1}{R_E} \frac{aR_E}{1+aR_E} \Delta V + \frac{a\frac{V_B}{2} + b}{1+aR_E}$$

Class AB subsystem: the current drain



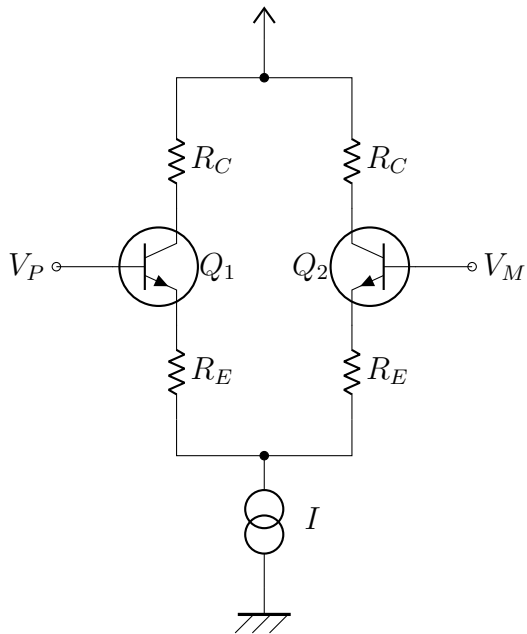
For the current drain subsystem:

Similarly, we get:

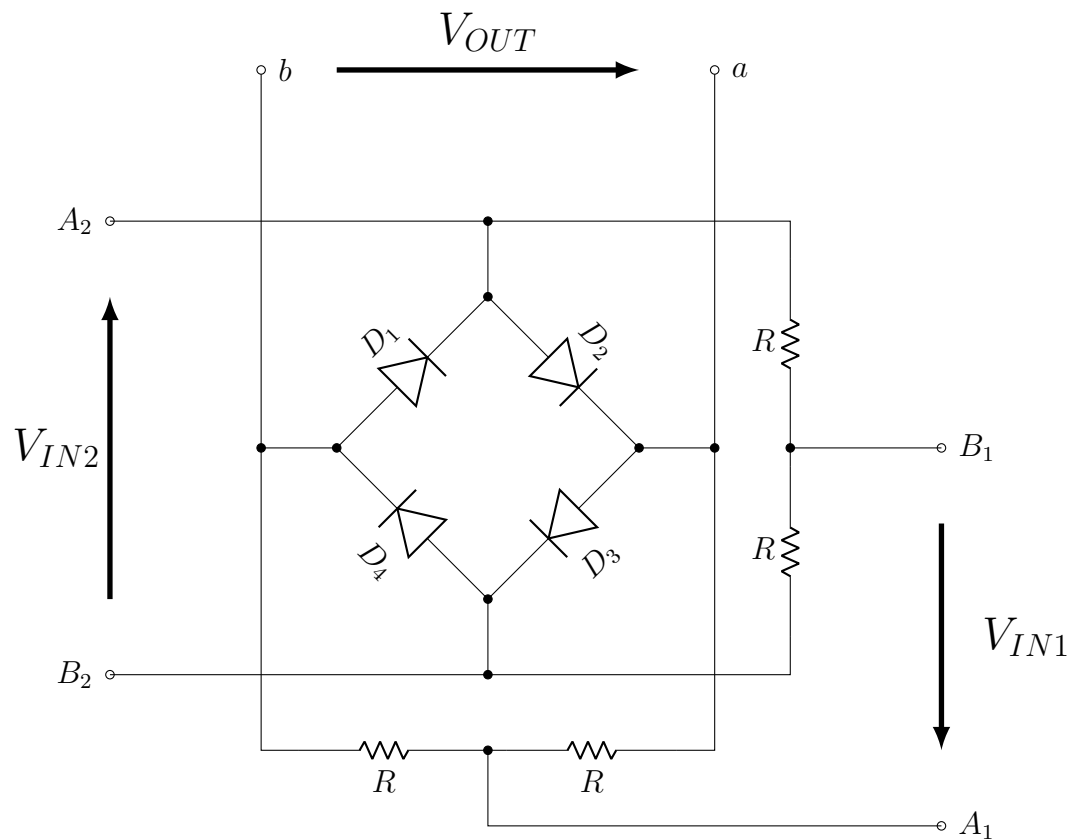
$$I_D = f_D \left(\Delta V + \frac{v_B}{2} - R_E I_D \right)$$

And:
$$I_D = \frac{1}{R_E} \frac{aR_E}{1+aR_E} \Delta V + \frac{a\frac{V_B}{2} + b}{1+aR_E}$$

Example 6: differential pair



Example 7: ring modulator



Example 8: the Gilbert Cell (analog multiplier)

That's a tough one, be patient...